Torchlette Sans

Aa Aa Aa Aa **Aa** **Aa** **Aa**

Aa *Aa* *Aa* *Aa* ***Aa*** ***Aa*** ***Aa***

We may say most aptly that the Analytical Engine weaves algebraical patterns just as the Jacquard-loom weaves flowers and leaves.

ADA LOVELACE · 1843

DISTINCTIVE FORMS

PROPORTIONS



ij!?

ROUND DOTS

oced

WIDE ROUND LETTERS

vwyzs

NARROW V W Y Z S

bdpq

LONG ASCENDERS AND DESCENDERS

0123456789

REDRAWN OLD-STYLE FIGURES

LOGISTIC SIGMOID

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

SOFTMAX

$$\operatorname{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

LINEAR LAYER

$$y = \sigma(Wx + b)$$

RECTIFIERS

$$\operatorname{ReLU}(x) = \max(0, x)$$
$$\operatorname{GELU}(x) = x\,\Phi(x)$$

MEAN SQUARED ERROR

$$\mathscr{L} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

CROSS-ENTROPY

$$\mathscr{L} = -\sum_k y_k \log \hat{y}_k$$

GRADIENT DESCENT

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathscr{L}$$

CHAIN RULE

$$\frac{\partial \mathscr{L}}{\partial w} = \frac{\partial \mathscr{L}}{\partial y} \frac{\partial y}{\partial w}$$

LEAST SQUARES

$$\hat{w} = (X^\top X)^{-1} X^\top y$$

RIDGE REGRESSION

$$\mathscr{L}(w) = \|y - Xw\|_2^2 + \lambda \|w\|_2^2$$

BAYES' RULE

$$p(\theta \mid x) = \frac{p(x \mid \theta)\, p(\theta)}{p(x)}$$

NORMAL DENSITY

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-(x-\mu)^2 / 2\sigma^2}$$

ENTROPY

$$H(p) = -\sum_x p(x) \log p(x)$$

KL DIVERGENCE

$$D_{\mathrm{KL}}(p \parallel q) = \sum_{x} p(x) \log \frac{p(x)}{q(x)}$$

SCALED DOT-PRODUCT ATTENTION

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\mathsf{T}}}{\sqrt{d_k}}\right)V$$

RESIDUAL CONNECTION

$$x_{\ell+1} = x_\ell + F_\ell(x_\ell)$$

LAYER NORMALIZATION

$$\mathrm{LN}(x) = \gamma \odot \frac{x - \mu}{\sqrt{\sigma^2 + \varepsilon}} + \beta$$

ADAM

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$
$$\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}$$

LOW-RANK ADAPTATION

$$W = W_0 + BA$$
$$B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times k}$$

EVIDENCE LOWER BOUND

$$\log p_\theta(x) \geq \mathbb{E}_{q_\varphi(z \mid x)}[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\varphi(z \mid x) \,\|\, p(z))$$

DIFFUSION, FORWARD STEP

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t \mathrm{I})$$

DIFFUSION, NOISE PREDICTION

$$\mathcal{L} = \mathbb{E}_{t, x_0, \varepsilon} \|\varepsilon - \varepsilon_\theta(x_t, t)\|^2$$

CONTRASTIVE LOSS, INFONCE

$$\mathcal{L} = -\log \frac{\exp(q \cdot k^+ / \tau)}{\sum_{i=0}^{K} \exp(q \cdot k_i / \tau)}$$

NOTATION IN ORDINARY TEXT

SIGNS AND INTERVALS

3–7 −3 ±2 ≈ 4

FRACTIONS AND ORDINALS

1st 2nd 3rd 4th ½ ¾ 5⁄16

RATIOS AND UNITS

1:2 10 ms 32 kg № 3

TENSOR DIMENSIONS

B×C×H×W y = A @ B

VERSION AND PROGRESS

v1.2.3 

CODE, UNITS AND TRANSFORMS

MUSIC IN TEXT

F♯ C♯ B♭ E♭

REARRANGEMENT

b C H W → B (H W) C

INFORMATION UNITS

8 bits 4 nats 2 Sh

NAMED OPERATORS

arg max arg min log sum exp

CHECKPOINTS

M20 HM45 ▶ 20

READABLE CODE ALTERNATIVES

CAPITAL I

I l 1 |

LOWERCASE L

l l 1 |

LATIN V AND GREEK NU

ν ν

ASCII SIGNS

≠ ≔ ↠ ⊢ ⊨ |→

TENSOR GRAMMAR



TENSOR LOOPS



NODES IN A MODEL

LATENT AND OBSERVED NODES



A CHAIN OF DEPENDENCIES



OBSERVED ENDPOINTS



$$A \setminus B \quad x \in \mathbb{R} \quad \mathscr{L} = \sum x^2$$

$$a \perp b \quad a \preccurlyeq b \quad a \succcurlyeq b$$

$$\Gamma \vdash \varphi \quad \Gamma \vDash \varphi \quad \Gamma \nvDash \psi$$

$$\Box p \Rightarrow \Diamond q \quad \vdash \Box p$$

$$[\![x]\!] \quad \langle \psi | \varphi \rangle \quad \hbar\omega$$

$$\blacksquare \quad \wr \quad \lightning \quad \ast \quad \blacktriangleright \quad \boxminus \quad \iota \quad \rho \quad \oslash \quad \phi \quad \boxslash$$

CONTEXTUAL SYMBOLS

MODAL NECESSITY

⊢□p □p □ Item a□b

SOLAR UNITS

M⊙ L⊙ R⊙

RATIO OPERATOR

a : b :: c : d

BRA-KET DELIMITERS

⟨ψ|φ⟩

RAW TENSOR PRIMITIVES

●○□⊢╵ₒ

LETTER JOINS AND WORDMARKS

LETTER JOINS

fb fh fk fj ft fft tf fy tw ry rw ll tp xt fu rv

WORDMARKS

Torchlette network gradient softmax Python

FURTHER WORDMARKS

backward forward output vector length residual

SIGMOID WORDMARK

sigmoid sigmoids

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefgij
klmnopqrstuvwxyzhıȷ

ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΘΣΤΥΦΧΨΩ∇αβγδεζη
θικλμνξοπρςστυφχψωϑϵϑϰϕϱϖ

𝕽 𝕹 𝖅 𝕼 𝕮 𝕻 𝕳 𝕬 𝕭 𝕯 𝕰 𝕱 𝕲

𝟘𝟙𝟚𝟛𝟜𝟝𝟞𝟟𝟠𝟡

𝓛 𝓑 𝓔 𝓕 𝓗 𝓘 𝓜 𝓡 𝒆 𝒈 𝒐

𝔄𝔅ℭ𝔇𝔈𝔉𝔊ℌℑ𝔍𝔎𝔏𝔐𝔑𝔒𝔓𝔔ℜ𝔖𝔗𝔘𝔙𝔚𝔛𝔜ℨ

𝒜ℬ𝒞𝒟ℰℱ𝒢ℋℐ𝒥𝒦ℒℳ𝒩𝒪𝒫𝒬ℛ𝒮𝒯𝒰𝒱
𝒲𝒳𝒴𝒵

𝑒 𝑔 𝑜

𝔄𝔅ℭ𝔇𝔈𝔉𝔊ℌℑ𝔍𝔎𝔏𝔐𝔑𝔒𝔓𝔔ℜ𝔖𝔗𝔘𝔙𝔚𝔛𝔜ℨ𝔞𝔟𝔠𝔡𝔢
𝔣𝔤𝔥𝔦𝔧𝔨𝔩𝔪𝔫𝔬𝔭𝔮𝔯𝔰𝔱𝔲𝔳𝔴𝔵𝔶𝔷

ABCDEFGHIJKLMNOPQRSTUVWXYZ

0123456789

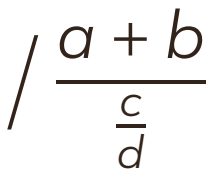
Thus, suppose a world to contain five things, α, β, γ, δ, ε. Then it will have a separate character for each of the 31 groups (with *non-existence* making up 32 or 2^5) shown in the following table:

TABLE II.

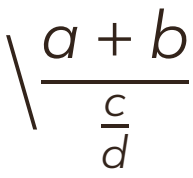
	αβ	αβγ	αβγδ	αβγδε
α	αγ	αβδ	αβγε	
β	αδ	αβε	αβδε	
γ	αε	αγδ	αγδε	
δ	βγ	αγε	βγδε	
ε	βδ	αδε		
	βε	βγδ		
	γδ	βγε		
	γε	βδε		
	δε	γδε		

If *A* stand for the smallest difference which *we perceive*, then we should have, instead of the formula Δ*s* = const., which is Fechner's, the formula Δ*s*/*s* = const., a formula which interprets all the *facts* of Weber's law, in an entirely different theoretic way from that adopted by Fechner.

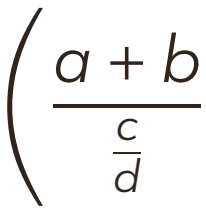
CAPITALS	ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩ
LOWERCASE	αβγδεζηθικλμνξοπρσςτυφχψω
SMALL CAPITALS	ABΓΔEZHΘIKΛMNΞOΠPΣTYΦXΨΩ
ITALIC	*ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩ* *αβγδεζηθικλμνξοπρσςτυφχψω*
TONOS & DIALYTIKA	Ά Έ Ή Ί Ό Ύ Ώ ά έ ή ί ό ύ ώ ϊ ϋ ΐ ΰ Ϊ Ϋ
MATHEMATICAL VARIANTS	ϑ ϕ ϖ ϱ ϵ



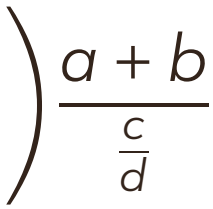
SLASH



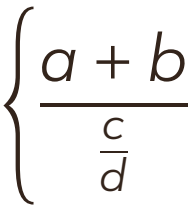
BACKSLASH



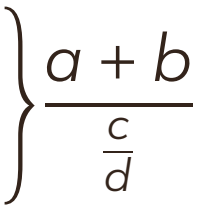
PARENLEFT



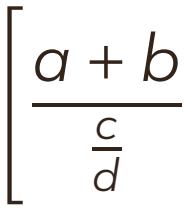
PARENRIGHT



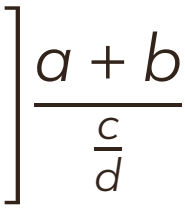
BRACELEFT



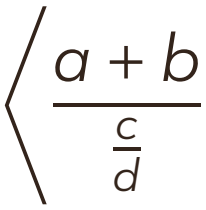
BRACERIGHT



BRACKETLEFT



BRACKETRIGHT



UNI27E8

PROPOSITION IV.

That axiom of metaphysicians which is termed the principle of contradiction, and which affirms that it is impossible for any being to possess a quality, and at the same time not to possess it, is a consequence of the fundamental law of thought, whose expression is $x^2 = x$.

Let us write this equation in the form

$$x - x^2 = 0,$$

whence we have

$$x(1 - x) = 0; \tag{1}$$

both these transformations being justified by the axiomatic laws of combination and transposition (ll. 13).

As y is, however, supposed to have the same meaning as x, we may replace it in the above equation by x, and we thus get

$$xx = x.$$

Now in common Algebra the combination xx is more briefly represented by x^2.

the latter is not interpretable, because the symbol -1 is not subject to the law $x(1-x)=0$, to which all class symbols are subject.

Let the following be the two equations, in which x and y are the unknown quantities:—

$$\begin{cases} mx + ny = d \\ m'x + n'y = d'. \end{cases}$$

We deduce $x = \dfrac{dn' - d'n}{n'm - nm'}$, and for y an analogous expression.

Number of the operations.	*Operation-cards* *Symbols indicating the nature of operations.*	*Cards of the variables.* *Columns on which operations are to be performed.*	*Columns which receive results of operations.*	*Progress of the operations.*
1	×	$V_2 \times V_4 =$	V_8	$= dn'$
2	×	$V_5 \times V_1 =$	V_9	$= d'n$
3	×	$V_4 \times V_0 =$	V_{10}	$= n'm$
4	×	$V_1 \times V_3 =$	V_{11}	$= nm'$
5	–	$V_8 - V_9 =$	V_{12}	$= dn' - d'n$
6	–	$V_{10} - V_{11} =$	V_{13}	$= n'm - nm'$
7	÷	$\dfrac{V_{12}}{V_{13}} =$	V_{14}	$= x = \dfrac{dn' - d'n}{n'm - nm'}$

INFERIOR FIGURES · SUBS

Again, it may be required, for example, to multiply an expression of the form $ax^m + bx^n$ by another $Ax^p + Bx^q$, and then to reduce the product to the least number of terms, if any of the indices are equal. The two factors being ordered with respect to x, the general result of the multiplication would be

$$Aax^{m+p} + Abx^{n+p} + Bax^{m+q} + Bbx^{n+q}.$$

So that if a problem require n substitutions, the successive series of numbers for the V's in the whole of them will be 2, 4, 8, 16 ... 2^n.

SUPERIOR LETTERS, FIGURES AND SIGNS · SUPS









1. ♘f3 1… ♞f6

M○ L○ R○

affirmation
FFI

shuffled
FFL

effect
FF

first
FI

flows
FL

attention
TT · SHARED CROSSBAR

The theory
DEFAULT

The theory
SS01 CAPITAL TH SS03 LOWERCASE TH

The theory of chance consists in reducing all the events of the same kind to a certain number of cases equally possible, that is to say, to such as we may be equally undecided about in regard to their existence, and in determining the number of cases favorable to the event whose probability is sought. The ratio of this number to that of all the cases possible is the measure of this probability, which is thus simply a fraction whose numerator is the number of favorable cases and whose denominator is the number of all the cases possible.

WORD MARKS · BACKWARD, FORWARD	A cylinder of soft steel, pressed with great force against the hardened steel engraving, is now made to roll very slowly backward and forward over it, thus receiving the design, but in relief.
WORD MARK · LENGTH	At Wheal Friendship mine in Cornwall, a different contrivance is employed: there is in that mine an inclined plane, passing underground about two-thirds of a mile in length.
WORD MARK · RESIDUAL	Now it may be, that no such residual relations exist.
WORD MARK · VECTOR	Mass, which used to be regarded as the most indubitable of physical quantities, is now generally believed to vary according to velocity, and to be, in fact, a vector quantity which at a given moment is different in different directions.
LOGOTYPE · NETWORK	Each of its associates becomes a hook to which it hangs, a means to fish it up by when sunk beneath the surface. Together, they form a network of attachments by which it is woven into the entire tissue of our thought.
THE SIGMOID I	Bechterew (Pflüger's Archiv, vol. 35, p. 137) found *no* anæsthesia in a cat with motor symptoms from ablation of sigmoid gyrus.
JOINS · TW FT XT FY FU	The twelfth chapter, which completes the first section, contains a few suggestions for the assistance of those who propose visiting manufactories. and having formed such conjectures, the desire to refute or to verify them, gave an additional interest to the pursuit.
JOINS · FH RW	The sense of my bodily existence, however obscurely recognized as such, *may* then be the absolute original of my conscious selfhood, the fundamental perception that *I am*. It was found that the domestic arrangement, or interior economy of factories, was so interwoven with the more general questions, that it was deemed unadvisable to separate the two subjects.

ft tw xt fy fu fh rw ry rv tp tf ll

PROSE LIGATURES · SS16

RANGE DASH · FRACTIONS	Dr. W. H. Burnham learned 16 lines of In Memoriam for 8 days; time, 14–17 minutes—daily average 14 ¾.
BUILT-UP FRACTIONS	One square yard of the substance of which these veils are made weighs 4 ⅓ grains, whilst one square yard of silk gauze weighs 137 grains, and one square yard of the finest patent net weighs 262 ½ grains.
OLD-STYLE PERCENT	Dr. Liébeault claims that he hypnotizes 92% of all comers, and Wetterstrand in Stockholm says that amongst 718 persons there proved to be only 18 whom he failed to influence.
RATIO COLON	With a blurred shadow in indirect vision the difference in favor of motion was much greater—namely, 13.3:40.7.
NUMERO	Bowditch and Southard in Journal of Physiology, vol. iii. № 3.
TUPLE SPACES	Here *f*(1,1) represents what *f*(*x*, *y*) becomes when we make therein *x*=1, *y*=1;
CAPITAL HYPHEN · TT	DESCENT FROM FLINT-WORKERS. RULES FOR LETTERING.

NOTATION FORMS

B×C×H×W

B×C×H×W

SHAPE

y = A @ B

y = A @ B

MATMUL

v1.2.3

v1.2.3

VERSION TAG

A => B <-> C

A ⇒ B ↔ C

ARROWS

{1,5,3,8}

SPARK BARS

{1,5,3,8}

SPARK LINE · SS15

[=======----]

PROGRESS BAR

STYLISTIC SETS

	DEFAULT	ALTERNATE
SS01 · CAPITAL TH	The theory	The theory
SS02 · ELLIPSIS	16 … 2^n	16 … 2^n
SS03 · LOWERCASE TH	the truths	the truths
SS04 · TWO-STOREY A	algebraical	algebraical
SS05 · Q	Q. E. D.	Q. E. D.
SS06 · Y	symbolical	symbolical
SS07 · T	the great	the great
SS08 · SEVEN	718 persons	718 persons

CHARACTER SET

CAPITALS

ABCDEFGHIJKLMNOPQRSTUVWXYZ

LOWERCASE

abcdefghijklmnopqrstuvwxyz

SMALL CAPITALS

ABCDEFGHIJKLMNOPQRSTUVWXYZ

ITALIC

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz

LINING FIGURES

0123456789

OLD-STYLE FIGURES

0123456789

SMALL-CAP FIGURES

0123456789

LIGATURES

ff fi fl ffi ffl tt

PUNCTUATION

.,:;...!¡?¿·•‘’“”,„‹›«»()[]{}/\|-–—_&@#*†‡§¶

SIGNS

+−±×÷=≠≈<>≤≥%‰°′″$¢£¥€№™©®

ACCENTED

ÀÁÂÃÄÅĀĂĄÆÇĆČĎĐÈÉÊËĒĖĘĚĞĢĪĮİĶ
ĹĻĽŁŃŅŇÑÒÓÔÕÖØŌŐŒŔŘŚŞŠŞŤŢÙÚ
ÛÜŪŮŰŲŴÝŸŶŹŻŽÞ

àáâãäåāăąæçćčďđèéêëēėęěğģīįıķĺļľłńņň
ñòóôõöøōőœŕřśşšşťţùúûüūůűųŵýÿŷźżžþð
ß

GREEK

ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩ

αβγδεζηθικλμνξοπρστυφχψω

WEIGHTS

LIGHT · 300

I think, therefore I am

REGULAR · 400

I think, therefore I am

MEDIUM · 500

I think, therefore I am

SEMIBOLD · 600

I think, therefore I am

BOLD · 700

I think, therefore I am

EXTRABOLD · 800

I think, therefore I am

BLACK · 900

I think, therefore I am

ITALICS

LIGHT ITALIC

Custom, then, is the great guide of human life.

ITALIC

Custom, then, is the great guide of human life.

MEDIUM ITALIC

Custom, then, is the great guide of human life.

SEMIBOLD ITALIC

Custom, then, is the great guide of human life.

BOLD ITALIC

Custom, then, is the great guide of human life.

EXTRABOLD ITALIC

Custom, then, is the great guide of human life.

BLACK ITALIC

Custom, then, is the great guide of human life.

ROMAN & ITALIC

LIGHT	The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.	*The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.*
REGULAR	The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.	*The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.*
MEDIUM	The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.	*The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.*
SEMIBOLD	The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.	*The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.*
BOLD	**The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.**	***The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.***
EXTRABOLD	**The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.**	***The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.***
BLACK	**The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.**	***The Analytical Engine has no pretensions whatever to originate any thing. It can do whatever we know how to order it to perform.***

An Investigation of the Laws of Thought

GEORGE BOOLE · 1854

The Principles of Psychology

WILLIAM JAMES · 1890

A Philosophical Essay on Probabilities

PIERRE-SIMON LAPLACE · 1814

Sketch of the Analytical Engine

L. F. MENABREA & ADA LOVELACE · 1843

An Enquiry Concerning Human Understanding

DAVID HUME · 1748

Passages from the Life of a Philosopher

CHARLES BABBAGE · 1864

Chance, Love, and Logic

CHARLES SANDERS PEIRCE · 1923

The Problems of Philosophy

BERTRAND RUSSELL · 1912

42 PT

Consciousness, then, does not appear to it-

30 PT

Consciousness, then, does not appear to itself chopped up in

22 PT

Consciousness, then, does not appear to it-self chopped up in bits. Such words as ‘chain’ or ‘train’ do not describe it fitly as it

16 PT

Consciousness, then, does not appear to itself chopped up in bits. Such words as ‘chain’ or ‘train’ do not describe it fitly as it presents itself in the first instance. It is nothing jointed;

12.5 PT

Consciousness, then, does not appear to itself chopped up in bits. Such words as ‘chain’ or ‘train’ do not describe it fitly as it presents itself in the first instance. It is nothing jointed; it flows. A ‘river’ or a ‘stream’ are the metaphors by which it is most naturally described. *In talking of it hereafter,*

10 PT

Consciousness, then, does not appear to itself chopped up in bits. Such words as ‘chain’ or ‘train’ do not describe it fitly as it presents itself in the first instance. It is nothing jointed; it flows. A ‘river’ or a ‘stream’ are the metaphors by which it is most naturally described. *In talking of it hereafter, let us call it the stream of thought, of consciousness, or of subjective life.*

8 PT

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6.5 PT

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IV

SECTION IV.

SCEPTICAL DOUBTS CONCERNING THE OPERATIONS OF THE UNDERSTANDING.

PART I.

20. All the objects of human reason or enquiry may naturally be divided into two kinds, to wit, *Relations of Ideas*, and *Matters of Fact*. Of the first kind are the sciences of Geometry, Algebra, and Arithmetic; and in short, every affirmation which is either intuitively or demonstratively certain. *That the square of the hypothenuse is equal to the square of the two sides*, is a proposition which expresses a relation between these figures. *That three times five is equal to the half of thirty*, expresses a relation between these numbers. Propositions of this kind are discoverable by the mere operation of thought, without dependence on what is anywhere existent in the universe. Though there never were a circle or triangle in nature, the truths demonstrated by Euclid would for ever retain their certainty and evidence.

21. Matters of fact, which are the second objects of human reason, are not ascertained in the same manner; nor is our evidence of their truth, however great, of a like nature with the foregoing. The contrary of every matter of fact is still possible; because it can never imply a contradiction, and is conceived by the mind with the same facility and distinctness, as if ever so conformable to reality. *That the sun will not rise to-morrow* is no less intelligible a proposition, and implies no more contradiction than the affirmation, *that it will rise*. We should in vain, therefore, attempt to demonstrate its falsehood. Were it demonstratively false, it would imply a contradiction, and could never be distinctly conceived by the mind.

It may, therefore, be a subject worthy of curiosity, to enquire what is the nature of that evidence which assures us of any real existence and matter of fact, beyond the present testimony of our senses, or the records of our memory. This part of philosophy, it is observable, has been little cultivated, either by the ancients or moderns; and therefore our doubts and errors, in the prosecution of so important an enquiry, may be the more excusable; while we march through such difficult paths without any guide or direction. They may even prove useful, by exciting curiosity, and destroying that implicit faith and security, which is the bane of all reasoning and free enquiry. The discovery of defects in the common philosophy, if any such there be, will not, I presume, be a discouragement, but rather an incitement, as is usual, to attempt something more full and satisfactory than has yet been proposed to the public.

22. All reasonings concerning matter of fact seem to be founded on the relation of *Cause and Effect*. By means of that relation alone we can go beyond the evidence of our memory and senses. If you were to ask a man, why he believes any matter of fact, which is absent; for instance, that his friend is in the country, or in France; he would give you a reason; and this reason would be some other fact; as a letter received from him, or the knowledge of his former resolutions and promises. A man finding a watch or any other machine in a desert island, would conclude that there had once been men in that island. All our reasonings concerning fact are of the same nature. And here it is constantly supposed that there is a connexion between the present fact and that which is inferred from it. Were there nothing to bind them together, the inference would be entirely precarious. The hearing of an articulate voice and rational discourse in the dark assures us of the presence of some person: Why? because these are the effects of the human make and fabric, and closely connected with it. If we anatomize all the other reasonings of this nature, we shall find that they are founded on the relation of cause and effect, and that this relation is either near or remote, direct or collateral. Heat and light are collateral effects of fire, and the one effect may justly be inferred from the other.

CHAPTER IX.

THE STREAM OF THOUGHT.

WE NOW BEGIN OUR STUDY OF THE MIND FROM WITHIN. Most books start with sensations, as the simplest mental facts, and proceed synthetically, constructing each higher stage from those below it. But this is abandoning the empirical method of investigation. No one ever had a simple sensation by itself. Consciousness, from our natal day, is of a teeming multiplicity of objects and relations, and what we call simple sensations are results of discriminative attention, pushed often to a very high degree. It is astonishing what havoc is wrought in psychology by admitting at the outset apparently innocent suppositions, that nevertheless contain a flaw. The bad consequences develop themselves later on, and are irremediable, being woven through the whole texture of the work. The notion that sensations, being the simplest things, are the first things to take up in psychology is one of these suppositions. The only thing which psychology has a right to postulate at the outset is the fact of thinking itself, and that must first be taken up and analyzed. If sensations then prove to be amongst the elements of the thinking, we shall be no worse off as respects them than if we had taken them for granted at the start.

FIVE CHARACTERS IN THOUGHT.

How does it go on? We notice immediately five important characters in the process, of which it shall be the duty of the present chapter to treat in a general way:

1) Every thought tends to be part of a personal consciousness.
2) Within each personal consciousness thought is always changing.
3) Within each personal consciousness thought is sensibly continuous.
4) It always appears to deal with objects independent of itself.
5) It is interested in some parts of these objects to the exclusion of others, and welcomes or rejects—*chooses* from among them, in a word—all the while.

. . .

Consciousness, then, does not appear to itself chopped up in bits. Such words as ‘chain’ or ‘train’ do not describe it fitly as it presents itself in the first instance. It is nothing jointed; it flows. A ‘river’ or a ‘stream’ are the metaphors by which it is most naturally described. *In talking of it hereafter, let us call it the stream of thought, of consciousness, or of subjective life.*

SMALL CAPITALS · SMCP + C2SC

PROPOSITION IV.

SMALL CAPITALS IN RUNNING TEXT

But immediately upon this I observed that, whilst I thus wished to think that all was false, it was absolutely necessary that I, who thus thought, should be somewhat; and as I observed that this truth, I think, therefore I am (COGITO ERGO SUM), was so certain and of such evidence that no ground of doubt, however extravagant, could be alleged by the sceptics capable of shaking it, I concluded that I might, without scruple, accept it as the first principle of the philosophy of which I was in search.

SMALL-CAPITAL HEADINGS WITH SMALL-CAPITAL FIGURES

CHAPTER IX. THE STREAM OF THOUGHT, 224

Consciousness tends to the personal form. It is in constant change. It is sensibly continuous. ‘Substantive’ and ‘transitive’ parts of Consciousness. Feelings of relation. Feelings of tendency. The ‘fringe’ of the object. The feeling of rational sequence. Thought possible in any kind of mental material. Thought and language. Consciousness is cognitive. The word Object. Every cognition is due to one integral pulse of thought. Diagrams of Thought’s stream. Thought is always selective.

GREEK SMALL CAPITALS

αβγδε ΑΒΓΔΕ FECHNER

SMALL CAPITALS IN LIGHT, MEDIUM, BOLD AND BLACK

THE PROBLEMS OF PHILOSOPHY

THE PROBLEMS OF PHILOSOPHY

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THE PROBLEMS OF PHILOSOPHY

THE PROBLEMS OF PHILOSOPHY

CAPITALS

PASSAGES

FROM

THE LIFE OF A PHILOSOPHER.

BY

CHARLES BABBAGE, ESQ., M.A.,

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MASSIL., ET DIVION., SOCIUS. ACAD. IMP. ET REG. PETROP., NEAP., BRUX.,
PATAV., GEORG. FLOREN, LYNCEI ROM., MUT., PHILOMATH. PARIS, SOC. CORR.,
ETC.

CAPITAL SPACING · CPSP CASE FORMS · CASE

(COGITO ERGO SUM)

DEFAULT

(COGITO ERGO SUM)

CASE + CPSP

PATAV., GEORG. FLOREN,

KERNING

1080 lbs

DEFAULT

1080 lbs

SLASHED ZERO · ZERO

OLD-STYLE · PROPORTIONAL · ONUM PNUM

1826214

Placing the most ancient epoch of history at five thousand years ago, or at 1826213 days, and the sun having risen constantly in the interval at each revolution of twenty-four hours, it is a bet of 1826214 to one that it will rise again to-morrow.

OLD-STYLE · TABULAR · ONUM TNUM

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LINING · PROPORTIONAL · LNUM PNUM

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LINING · TABULAR · LNUM TNUM

1826214

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SMALL-CAPITAL FIGURES · C2SC

43949268

SLASHED ZERO · ZERO

4005

ALTERNATE SEVEN · SS08

11748

Thus in the lottery of France, formed as is known of 90 numbers of which five are drawn at each draw, the probability of drawing a given combination is 5/90, or 1/18; the lottery ought then for the equality of the play to give eighteen times the stake. The total number of combinations two by two of the 90 numbers is 4005, and that of the combinations two by two of 5 numbers is 10. The probability of the drawing of a given pair is then 1/4005, and the lottery ought to give four hundred and a half times the stake; it ought to give 11748 times for a given tray, 511038 times for a quaternary, and 43949268 times for a quint.

The new series thus formed by the addition of his own triangular numbers is—

Number.	*Table.*	*1st Difference.*	*2nd Difference.*	*3rd Difference.*
1	1	3	3	1
2	4	6	4	1
3	10	10	5	1
4	20	15	6	
5	35	21		
6	56			

He will at once perceive that this Table of the number of cannon balls contained in a triangular pyramid can be carried to any extent by simply adding successive differences, the third of which is constant.

TABULAR LINING FIGURES · LNUM TNUM, WITH ORDINALS

Now the Table we have been considering is, in fact, merely a Table whose first difference is constant and equal to five. If we express it in pence it becomes—

	Table.	*1st Difference.*
1	5	5
2	10	5
3	15	5
4	20	5
5	25	

TABULAR OLD-STYLE FIGURES · ONUM TNUM

Now an inquisitive boy would naturally count the numbers in each group and he would find them thus—

1 3 6 10 15 21

PROPORTIONAL OLD-STYLE FIGURES · ONUM PNUM

Thus 36 is the product of 6 multiplied by 6, and 36 is called the square of 6. I would now recommend him to examine the series of square numbers

1, 4, 9, 16, 25, 36, 49, 64, &c.,

PROPORTIONAL LINING FIGURES · LNUM PNUM

A block of squared stone was taken for the subject of experiment:

1.	Weight of stone	1080 lbs
2.	In order to drag this stone along the floor of the quarry, roughly chiselled, it required a force equal to	758 lbs
3.	The same stone dragged over a floor of planks required	652 lbs
4.	The same stone placed on a platform of wood, and dragged over a floor of planks, required	606 lbs
5.	After soaping the two surfaces of wood which slid over each other, it required	182 lbs
6.	The same stone was now placed upon rollers of three inches diameter, when it required to put it in motion along the floor of the quarry	34 lbs
7.	To drag it by these rollers over a wooden floor	28 lbs
8.	When the stone was mounted on a wooden platform, and the same rollers placed between that and a plank floor, it required	22 lbs

From this experiment it results, that the force necessary to move a stone along

	Part of its weight
The roughly chiselled floor of its quarry is nearly	2/3
Along a wooden floor	3/5
By wood upon wood	5/9
If the wooden surfaces are soaped	1/6
With rollers on the floor of the quarry	1/32
On rollers on wood	1/40
On rollers between wood	1/50

The probability of the drawing of a given pair is then 1/4005, and the lottery ought to give four hundred and a half times the stake; it ought to give 11748 times for a given tray, 511038 times for a quaternary, and 43949268 times for a quint.

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⁰⁄₃

$\frac{1}{2}$ $\frac{12}{345}$ $\frac{1234}{5678}$

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SIGNS

DOUBLE-STRUCK CAPITALS

ℕ ℤ ℚ ℝ ℂ

ARROWS

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PROSE ARROWS AND SIGNS FROM PLAIN KEYS · SS02

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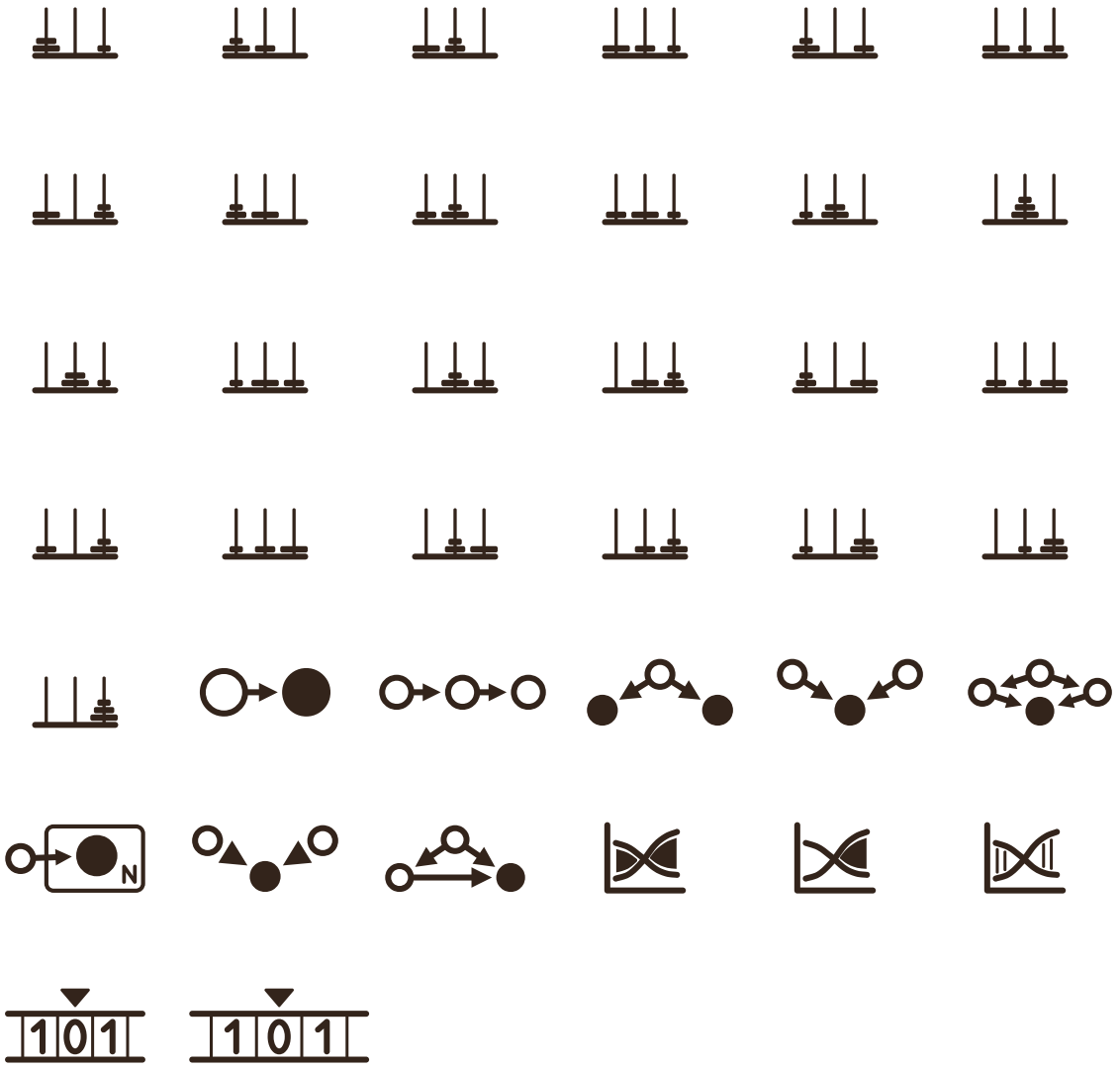
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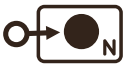
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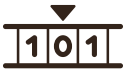
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